VS-05 Statistical Product Composition

Table of contents

1. Purpose and value created . . . 1
2. Position and state transition . . . 2
3. Stages . . . 3
    - VS-05.1 Select and Integrate Source Datasets . . . 3
    - VS-05.2 Construct Composite Indicators or Indices . . . 4
    - VS-05.3 Perform Consistency and Coherence Checks . . . 4
    - VS-05.4 Generate Analytical and Interpretive Outputs . . . 5
    - VS-05.5 Approve and Register Statistical Products . . . 5
4. The four process types . . . 6
    - Design: provides metadata . . . 6
    - Implementation: provides the execution process . . . 9
    - Execution: provides the statistical data . . . 12
    - Management: provides quality reports . . . 13
    - Metadata and paradata per process type . . . 16
5. Actors and roles . . . 17
6. Capabilities, processes, and functions . . . 18
    - Capability foundations . . . 18
    - Stages, processes, and functions . . . 19
    - Framework mappings . . . 19
7. Principles . . . 20
8. Interfaces, services, and events . . . 21
9. Key performance indicators . . . 23
10. Toolkit . . . 24

1. Purpose and value created

VS-05 turns separate analytical results into finished statistical products. It takes validated statistics from VS-04 and brings them together into coherent products: integrated tables, composite indicators, indices, time series, and analytical publications with their narrative and visualizations. VS-05 does not re-estimate. It composes validated outputs as they are received, and it carries their accuracy and methodology into the product. Every product is checked for internal consistency and coherence with related statistics, explained in plain language, and approved against dissemination and confidentiality standards. The value created is **coherence and interpretability**. Users receive a product that is consistent, understandable, and signed off by the NSI.

VS-05 corresponds to the business domain **Statistically composed and composite products** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Subject-matter statistician, author, or release manager:** the stages in attribute 3, which describe what happens in each production cycle.
- **Statistical designer or methodologist:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.

- **Manager who wants the big picture:** attributes 1, 2, and 8. Examples and end-to-end scenarios are part of the SAF Toolkit.

> **i** Signature discipline: coherence and interpretability
>
> VS-05 composes; it does not re-estimate. Composite products are internally consistent, methodologically sound, coherent with related and earlier statistics, and accompanied by the metadata and interpretation that users need. Composition never silently changes a component estimate. Any adjustment for coherence is documented and traceable. Every product carries its lineage to the component outputs and the composition method, and it is approved before it is registered.

> **i** Metadata inherited from VS-04
>
> VS-05 receives Statistics together with the metadata VS-04 attached: the methodology, the weights, the accuracy measures, and the record of the disclosure control already applied. VS-05 carries this accuracy and quality metadata into the composite product. It does not discard or recompute it. It uses the metadata to weight or caveat composed figures, and to run coherence checks across the combined outputs.

PLACEHOLDER: ArchiMate view VS-05 (a) value stream and capabilities

2. Position and state transition

VS-05 is the fifth primary value stream. It sits between estimation (VS-04) and dissemination (VS-06). It receives Statistics when VS-04 publishes the *Output Data Registered* event, and it hands over approved statistical products to VS-06.

VS-05 works within State #4. Before VS-05, the data is a set of separate versions of Statistics with the status *analytical*. After VS-05, it is a registered, immutable product version with the status *approved*. That version refers to the analytical versions it was composed from, so approval never changes an analytical version. The product is coherent across sources and time, explained, checked against dissemination and confidentiality standards, and signed off. The two gate moments, the minimum metadata set, and the protection of State #4 are described in *State #4: Statistics* and are not repeated here. **[VC22]**

A product can draw on analytical versions from several domains, surveys, and periods. Composition then often reveals incoherence that no single domain could see. In that case the discrepancy is explained or the component goes back to VS-04; it is never silently adjusted in VS-05.

From	To	Gate	Event out
#4 Statistics (status *analytical*)	#4 Statistics (status *approved*)	Approval gate of State #4 (component lineage, propagated accuracy, internal consistency, coherence, interpretive content, confidentiality including combination risk, approval record)	*Statistical Product Registered*

PLACEHOLDER: ArchiMate view VS-05 (c) state handover and events: analytical Statistics (#4) from several domains, the composition service (selection and integration, composite construction, coherence checks, interpretive content, approval), approved Statistics (#4) inside the composition domain

3. Stages

The stages describe the value that VS-05 adds in every production cycle. Together they form the execution process type of VS-05. In a mature steady state, the product specifications, the composite indicators, and the coherence checks have been designed and built in advance (see attribute 4). Execution then focuses on composing the products, checking coherence, writing the content, and obtaining approval.

Stage	Name	Value added
VS-05.1	Select and Integrate Source Datasets	The right validated outputs are brought together as the basis for a coherent product
VS-05.2	Construct Composite Indicators or Indices	Indices, ratios, and time series summarize the data and add meaning
VS-05.3	Perform Consistency and Coherence Checks	The product is internally consistent and fits the wider statistical picture
VS-05.4	Generate Analytical and Interpretive Outputs	Users can understand the product through narrative, key messages, and visualizations
VS-05.5	Approve and Register Statistical Products	An approved product is registered and ready for dissemination (State #4, *approved*)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-05.1 Select and Integrate Source Datasets

Value created: *The right validated outputs brought together as the basis for a coherent product.*

The stage starts when VS-04 publishes the *Output Data Registered* event for a component that a product needs, or when all components for a scheduled release are available. The designed selection rules pick the correct analytical versions of State #4 by domain, period, and accuracy. The components are integrated into the product structure. Their accuracy and methodology metadata is carried forward into the product. The exact component versions used are recorded, so the product can be reproduced later. Components from several domains, surveys, or periods are combined according to the product specification.

Activity	Key outputs	SAF process	Other reference
Select the correct component versions according to the product specification	Selected components	PRD-6.1 Prepare draft outputs; DLM-6 Provide data for reuse	SAF-P08 Data is shared property
Integrate the components into the product structure	Integrated product base	PRD-6.1 Prepare draft outputs	1.3.6 Integrate statistics
Carry component accuracy and methodology metadata into the product	Propagated metadata	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Record component lineage: which versions were used	Component lineage	OVA-2 Metadata Management	CSDA Traceability

VS-05.2 Construct Composite Indicators or Indices

Value created: *Derived measures, such as indices, ratios, and time series, that summarize the data and add meaning.*

This stage computes the designed composite indicators, indices, and time-series aggregates from the integrated components. Indicators and indices are calculated with the designed formulas and weights. Time series are chained, linked, and benchmarked according to the specification. Each composite receives an uncertainty measure derived from the accuracy of its components. No component estimate is re-estimated; composites are built from the validated outputs as received. The composition method and its inputs are recorded in the lineage. Products without composites skip this stage, and the reason is recorded.

Activity	Key outputs	SAF process	Other reference
Calculate composite indicators and indices with the designed formulas	Composite indicators	PRD-6.1 Prepare draft outputs	1.3.6 Integrate statistics
Construct time series: chain, link, benchmark	Composite time series	PRD-6.1 Prepare draft outputs	
Derive the uncertainty of composites from component accuracy	Composite accuracy measures	PRD-6.1 Prepare draft outputs	2.1.3 Statistical quality management
Record the composition method and inputs as lineage	Composite lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

VS-05.3 Perform Consistency and Coherence Checks

Value created: *Confidence that the product is internally consistent and fits the wider statistical picture.*

This stage runs the designed checks. Internal consistency checks confirm that totals reconcile with their components and that breakdowns add up to totals. Coherence checks compare the product with related statistics, such as the national accounts, and with earlier releases, within the designed tolerances. Each discrepancy receives a disposition according to policy: accept, flag, or return for review. Significant discrepancies are explained. No silent adjustment is made; any reconciliation is documented and traceable. A discrepancy that points to a component error goes back to VS-04.

Activity	Key outputs	SAF process	Other reference
Run internal consistency checks: totals and breakdowns reconcile	Consistency results	PRD-6.2 Validate outputs	
Run coherence checks against related and earlier statistics	Coherence results	PRD-6.2 Validate outputs	1.3.6 Integrate statistics
Assign a disposition to discrepancies (accept, flag, return) and explain them	Discrepancy dispositions	PRD-6.2 Validate outputs	GAMSO Quality Management
Log all check results in the product audit trail	Audit trail entries	OVA-4 Process Data Management	SAF-P10 Security by design

PLACEHOLDER: VS-05 coherence check flow: integrated components and composites passing internal consistency and coherence checks against related and earlier statistics; discrepancies accepted, flagged with an explanation, or returned to VS-04; all results written to the product audit trail

VS-05.4 Generate Analytical and Interpretive Outputs

Value created: *Products that users can understand: narrative, key messages, and visualizations grounded in the data.*

This stage produces the content that people read. The analytical narrative and commentary are written according to the templates. Key messages are stated together with the evidence that supports them. Tables and visualizations are produced. The content goes through the editorial and quality review workflow, which checks that it is accurate, written in plain language, accessible, and consistent with the data and its caveats. The accuracy carried from VS-04 is reflected in the wording: a change within the margin of error is not presented as a real change.

Activity	Key outputs	SAF process	Other reference
Write the analytical narrative and commentary according to the templates	Product narrative	PRD-6.3 Interpret and explain outputs	1.4.1 Statistical (output) analysis
State key messages with their supporting evidence	Key messages	PRD-6.3 Interpret and explain outputs	
Produce tables and visualizations	Tables and visualizations	PRD-7.2 Produce dissemination products	
Run the editorial and quality review of the content	Reviewed content	OVA-1 Quality Management	GAMSO Quality Management

VS-05.5 Approve and Register Statistical Products

Value created: *A validated, approved, registered product, ready for dissemination.*

This stage validates, approves, and registers the product. The product is checked against the dissemination and confidentiality standards. This includes the disclosure risk that arises from combining outputs, because figures that are safe on their own can reveal confidential data when they are combined. The product is then submitted to the designated approval authority. On approval, the data passes the approval gate of State #4, and a persistent, immutable product version with the status *approved* is created. It carries a unique identifier and full lineage: components, method, coherence results, and the approval record. It is registered in the data catalog, and access permissions are applied. The stage ends by publishing the *Statistical Product Registered* event, which tells VS-06 that an approved product is available.

Output: Approved statistical product (State #4, status *approved*), the handover artifact to the dissemination domain. Its structure and minimum metadata set are described in *State #4: Statistics*.

Activity	Key outputs	SAF process	Other reference
Check the product against dissemination and confidentiality standards, including combination risk	Validation result	PRD-6.4 Apply disclosure control; PRD-6.5 Finalize outputs	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure

Activity	Key outputs	SAF process	Other reference
Submit the product to the approval authority and obtain sign-off	Approval record	PRD-6.5 Finalize outputs; DLM-2 Promote data through a quality gate	4.1.5 Accountability
Create an immutable, identified product version with full lineage (status *approved*)	Versioned product	DLM-5 Store and version data	CSDA Traceability
Register metadata and apply access permissions	Catalog registration, access-controlled product	DLM-3 Register and catalog data; DLM-4 Classify and protect data	SAF-P10 Security by design
Publish the *Statistical Product Registered* event to notify VS-06 Dissemination and Service Delivery	Event notification	DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

PLACEHOLDER: VS-05 product version structure: the approved product version linked to its component versions, composition method, propagated accuracy, coherence results, interpretive content, and approval record

4. The four process types

VS-05 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the statistical data, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-05. Their names describe the work in the context of product composition. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-05 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. In VS-05 these are the product specifications, the composite-indicator definitions, the coherence checks, the content templates, and the approval and confidentiality policy. Without them, composition becomes ad hoc and incoherent, and management has no baseline to measure coherence and interpretability against. Design runs when a product is introduced or changed, when an index is rebased, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design the Statistical Product Portfolio	PRD-1.3 Establish output objectives; PRD-2.1 Design outputs and dissemination	Product specifications, component selection rules, metadata propagation rules
Design Composite Indicators and Indices	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis	Composite-indicator specification, time-series specification, uncertainty representation

Design sub-process	Refines	Main outputs
Design Consistency and Coherence Checks	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Consistency and coherence check sets, tolerance policy, revision policy
Design Analytical and Interpretive Content	PRD-2.1 Design outputs and dissemination	Content templates, visualization specification, key-message guidelines, editorial review policy
Design Approval, Confidentiality, and Versioning Framework	PRD-2.1 Design outputs and dissemination; DLM-1 Plan and design data assets; DLM-4 Classify and protect data	Approval policy, confidentiality standard, versioning and lineage policy, access control design

Design the Statistical Product Portfolio

Refines: PRD-1.3 Establish output objectives; PRD-2.1 Design outputs and dissemination.

The starting point is the definition of what products will be produced. This sub-process specifies each statistical product: its purpose, target users, content, structure, breakdowns, and release frequency. It defines the analytical versions of State #4 that each product draws on, and the rules for selecting them by domain, period, and accuracy. It also defines how the accuracy and quality metadata of the components is carried into the product. Each product receives its classification and versioning identity. Reuse of existing validated outputs comes before new analysis, in line with SAF-P08.

Activity	Key outputs	SAF process	Other reference
Specify each product: purpose, users, content, structure, breakdowns, frequency	Product specification	PRD-1.3 Establish output objectives; PRD-2.1 Design outputs and dissemination	1.1.2 Design statistics product
Define the component outputs each product draws on and the selection rules	Component selection specification	PRD-2.1 Design outputs and dissemination	SAF-P08 Data is shared property
Define how component accuracy and quality metadata is carried into the product	Metadata propagation rules	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification
Define the product classification and versioning identity	Product metadata standard	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification

Design Composite Indicators and Indices

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis.

This sub-process specifies the derived measures that products present: composite indicators, indices, ratios, and time-series aggregates. Each is defined by its formula, its component inputs, its weighting or aggregation method, and its base period. For time series, the rules for chaining, linking, and benchmarking are specified. Where a composite combines outputs of different accuracy, the design specifies how its uncertainty is represented. Each composite receives its metadata and classification. International manuals, such as those for price and production indices, are followed where they apply.

Activity	Key outputs	SAF process	Other reference
Specify composite indicators and indices: formula, inputs, weighting, base period	Composite-indicator specification	PRD-2.5 Design processing and analysis	1.3.6 Integrate statistics

Activity	Key outputs	SAF process	Other reference
Specify the construction of time series: chaining, linking, benchmarking	Time-series specification	PRD-2.5 Design processing and analysis	
Define how component uncertainty is represented in composites	Uncertainty representation rules	PRD-2.5 Design processing and analysis	SAF-P09 No data without metadata and classification
Define metadata and classification for each composite	Composite metadata standard	PRD-2.1 Design outputs and dissemination	SAF-P09 No data without metadata and classification

Design Consistency and Coherence Checks

Refines: PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

This sub-process designs the checks that make sure integrated products are sound. It specifies internal consistency checks: totals reconcile with their components, and breakdowns add up to totals. It specifies coherence checks against related statistics, such as the national accounts or earlier products, and across time. It sets the tolerance thresholds and the disposition rules: accept, flag, or return for review. It also sets the revision policy for products. Together, these rules make up the coherence part of the approval gate of State #4.

Activity	Key outputs	SAF process	Other reference
Define internal consistency checks: totals and breakdowns reconcile	Consistency check set	PRD-2.5 Design processing and analysis	
Define coherence checks against related statistics and across time	Coherence check set	OVA-1 Quality Management	GAMSO Quality Management
Define tolerance thresholds and disposition rules	Coherence tolerance policy	PRD-2.5 Design processing and analysis	
Define the revision policy for products	Revision policy	OVA-1 Quality Management	GAMSO Quality Management

Design Analytical and Interpretive Content

Refines: PRD-2.1 Design outputs and dissemination.

This sub-process designs the content that makes products understandable. It specifies the structure of the analytical narrative and commentary. It specifies the tables and visualizations. It sets the guidelines for key messages and the evidence each message must rest on. It designs the editorial and quality review workflow for interpretive content. Standards for accessibility and plain language are designed in from the start, so that the product can be used by the public as well as by experts.

Activity	Key outputs	SAF process	Other reference
Specify the structure of the analytical narrative and commentary	Content template	PRD-2.1 Design outputs and dissemination	1.4.1 Statistical (output) analysis
Specify tables and visualizations	Visualization specification	PRD-2.1 Design outputs and dissemination	
Define key messages and the evidence they require	Key-message guidelines	PRD-2.1 Design outputs and dissemination	

Activity	Key outputs	SAF process	Other reference
Define the editorial and quality review workflow, including accessibility and plain language	Editorial review policy	PRD-2.1 Design outputs and dissemination	GAMSO Quality Management

Design Approval, Confidentiality, and Versioning Framework

Refines: PRD-2.1 Design outputs and dissemination; DLM-1 Plan and design data assets; DLM-4 Classify and protect data.

This last design sub-process makes sure that products are approved, protected, and traceable. It defines the approval workflow and the authority that signs off each type of product. It defines the confidentiality standards that a product must meet before registration, including the risk that arises from combining outputs. It defines the versioning policy for products: identification, triggers for revision, immutability, and retention. It defines the lineage from each product to its component versions. Role-based access controls are designed in line with SAF-P10, because approved products remain confidential until their release.

Activity	Key outputs	SAF process	Other reference
Define the approval workflow and authority	Approval policy	PRD-2.1 Design outputs and dissemination	4.1.5 Accountability
Define confidentiality standards for products, including combination risk	Confidentiality standard	DLM-4 Classify and protect data	SAF-P11 Privacy by design; 1.3.7 Statistical disclosure
Define the versioning and lineage requirements for products	Versioning and lineage policy	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control for products before release	Access control design	DLM-4 Classify and protect data	SAF-P10 Security by design; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-05 design process decomposition: inputs (user needs, analytical versions of State #4, standards and index manuals, confidentiality rules), the five design sub-processes in order, and the design artifact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the composition pipeline until it is ready for production. Nothing runs operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01. Index calculation and visualization tools shared in the NSI community are good candidates for reuse.

Implementation sub-process	Refines	Main outputs
Build Integration and Composition Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Integration engine, composition engine, version-fixed connection to the catalog
Build Indicator-Construction and Time-Series Engines	PRD-3.2 Reuse or build processing and analysis components	Indicator engine, time-series engine, composite accuracy component

Implementation sub-process	Refines	Main outputs
Build Coherence-Check, Content, and Approval Tooling	PRD-3.2 Reuse or build processing and analysis components; PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data	Coherence tooling, content tooling, approval and versioning store, events
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, coherence and reproducibility reports, production-ready pipeline

Build Integration and Composition Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

This sub-process turns the product designs into operational tools. The integration engine is built or configured to select and combine the designed component outputs into the product structure, together with their accuracy and quality metadata. The composition engine is built to assemble products from their components according to the specifications. Connections to the catalog of State #4 are configured with fixed versions, so that the exact component versions used are recorded. All composition logic is version-controlled in line with SAF-P10. Any product can therefore be reproduced from its component versions and the recorded method.

Activity	Key outputs	SAF process	Other reference
Build or configure the integration engine: select and combine component outputs	Configured integration engine	PRD-3.2 Reuse or build processing and analysis components	1.3.6 Integrate statistics
Build or configure the composition engine: assemble products from components	Configured composition engine	PRD-3.2 Reuse or build processing and analysis components	
Configure version-fixed connections to the catalog of State #4	Integration with the output catalog	PRD-3.4 Configure workflows	CSDA Metadata Management
Put all composition logic under version control	Versioned composition repository	PRD-3.4 Configure workflows	SAF-P10 Security by design

Build Indicator-Construction and Time-Series Engines

Refines: PRD-3.2 Reuse or build processing and analysis components.

In this sub-process the designed composite indicators and time series become automated services. Calculators for indicators and indices are built from the designed formulas and weights. Time-series engines are configured for chaining, linking, and benchmarking. The representation of uncertainty is implemented, so that every composite carries an appropriate accuracy measure. Every composite output is produced with its method and inputs attached. Tests on historical data confirm that the engines reproduce existing published series before they replace older tools.

Activity	Key outputs	SAF process	Other reference
Build calculators for indicators and indices from the designed formulas	Indicator construction engine	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Configure chaining, linking, and benchmarking of time series	Time-series engine	PRD-3.2 Reuse or build processing and analysis components	
Implement the representation of uncertainty for composites	Composite accuracy component	PRD-3.2 Reuse or build processing and analysis components	SAF-P09 No data without metadata and classification
Attach method and inputs to every composite	Binding of composite metadata	PRD-3.2 Reuse or build processing and analysis components	SAF-P09 No data without metadata and classification

Build Coherence-Check, Content, and Approval Tooling

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data.

This sub-process implements the coherence checks, the tooling for content, and the approval and versioning store. Coherence tooling implements the designed consistency checks and the checks across sources and time, with their tolerances. Content tooling supports the narrative, tables, and visualizations according to the templates, with the editorial review workflow. The approval and versioning store is built so that each approved product version is immutable, identified, reproducible, and accompanied by its lineage and approval record. Event notifications are configured, so that VS-06 is alerted when a product is approved and registered.

Activity	Key outputs	SAF process	Other reference
Implement consistency and coherence checks with their tolerances	Coherence tooling	PRD-3.2 Reuse or build processing and analysis components	
Build content and visualization tooling with the editorial workflow	Content tooling	PRD-3.3 Reuse or build dissemination components	
Build the approval and versioning store: immutable, identified, reproducible	Approval and versioning store	DLM-5 Store and version data	CSDA Traceability
Configure the *Statistical Product Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows; DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the pipeline goes into production, it must be thoroughly verified. Technical testing covers all components: integration, composition, indicator and time-series engines, coherence checks, content tooling,

and the approval store. Integration testing checks the end-to-end flow from the *Output Data Registered* event to the registration of an approved product. A coherence test confirms that composites reconcile with their components and fit earlier products. A reproducibility test confirms that a product can be regenerated identically from its component versions and the composition method. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from output intake to registration	Integration test report	PRD-3.5 Test production systems	
Run the coherence test: composites reconcile with components and earlier products	Coherence test report	PRD-3.6 Test statistical business process	
Run the reproducibility test: regenerate a product from components and method	Reproducibility report	PRD-3.6 Test statistical business process	CSDA Traceability
Finalize the production system and document operational procedures	Operational procedures manual, production-ready pipeline	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-05 implementation pipeline: the five design artifacts feeding the three build sub-processes, followed by technical testing, coherence testing, and reproducibility testing, resulting in a production-ready composition pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-05 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when all components for a scheduled release have been registered by VS-04, or when a revision of a published product is due. The stages run in order, with a loop back to VS-05.1 or VS-04 when a coherence check fails. Execution applies the designs and the solution as they are; a new product or a change in an index base goes back to Design. Every run ends in one of three outcomes: approved and registered, returned to VS-04 for a component, or held by the approval authority.

Aspect	VS-05
Trigger	All components of a scheduled release registered (*Output Data Registered*); a scheduled revision
Cadence	Per release, according to the release calendar
Performed by	ROL-15 Subject-matter statistician, ROL-10 Data engineer, ROL-17 Release manager, ROL-03 Data steward
Input	Analytical versions of Statistics (#4), product specifications, content templates
Output	Registered approved product version (#4, *approved*), its lineage, and its approval record
Control point	Approval gate of State #4 in VS-05.5 Approve and Register Statistical Products

Aspect	VS-05
Exception path	Coherence failure back to VS-04 Estimation, Analysis, and Statistical Modeling; combination risk above the threshold to ROL-16 Disclosure control specialist; refusal by the approval authority back to VS-05.3 Perform Consistency and Coherence Checks or VS-05.4 Generate Analytical and Interpretive Outputs

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the composition pipeline, assesses the coherence and interpretability of products, makes sure that every product is approved and can be reproduced, and feeds improvements back into Design. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, throughput report, paradata records
Assess Product Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Coherence report, interpretability review, completeness report, trend analysis
Manage Product and Version Lifecycle	OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use	Product version register, metadata quality report, version store audit
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Reproducibility audit, approval and confidentiality reports to VS-07 Governance and Compliance
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-05 must be tracked systematically across production cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9). They monitor throughput against the release calendar, because a late product delays publication. They track paradata: which component versions and methods were used, when, by whom, and how many approval rounds were needed. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per production cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Monitor throughput against the release calendar	Throughput report	OVA-4 Process Data Management	

Activity	Key outputs	SAF process	Other reference
Track paradata: component versions, methods, approval rounds	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Product Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the products is evaluated against the criteria from Design. Coherence covers internal consistency and consistency with related and earlier statistics. Interpretability covers the clarity and correctness of the narrative and visualizations. Completeness checks that all designed content is present. Accuracy propagation checks that the accuracy of components was carried correctly into the product. Quality indicators are produced per product and per cycle. Trends are tracked to detect recurring coherence problems or weaknesses in interpretation. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate coherence: internal, and with related and earlier statistics	Coherence report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess the interpretability of narrative and visualizations	Interpretability review	OVA-1 Quality Management	CSDA Data Quality
Check completeness and correct propagation of accuracy	Product completeness report	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across products and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality

Manage Product and Version Lifecycle

Refines: OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use.

The product portfolio and the versions it produces must stay current, complete, and well governed. This sub-process governs the life cycle of product specifications, composite-indicator definitions, and content templates. It versions them, manages transitions such as the rebasing of an index, and retires products that are no longer needed. It verifies that every product carries its full component lineage, composition method, coherence results, and approval record, as a compliance check on SAF-P09. It also governs the versioning store: retention, immutability, revisions, and the reconciliation of catalog entries with stored versions. Lineage from analytical versions through composition to approved products is tracked.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of products, indicators, and templates	Product version register	OVA-2 Metadata Management	SAF-P03 Active life cycle management

Activity	Key outputs	SAF process	Other reference
Verify metadata completeness: component lineage, method, coherence, approval	Metadata quality report	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Govern the versioning store: retention, immutability, revisions, reconciliation	Version store audit	DLM-5 Store and version data; DLM-7 Revise data	CSDA Traceability
Track metadata lineage from analytical versions to approved products	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use.

This sub-process produces the evidence that VS-05 operates within methodological, legal, and organizational requirements. Reproducibility audit reports show that every product can be regenerated from its component versions and the composition method. Approval and confidentiality reporting confirms that every registered product was approved by the designated authority and meets the confidentiality standard, including the risk of combining outputs. Standards compliance reporting confirms conformance to product and exchange standards, such as SDMX. Security compliance reporting verifies the access controls and the integrity of the approval record. These reports meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce reproducibility audit reports: every product can be regenerated	Reproducibility audit summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability
Report on approval and confidentiality compliance of products	Approval and confidentiality report	OVA-1 Quality Management	SAF-P11 Privacy by design
Report on standards compliance: products, SDMX exchange	Standards compliance report	OVA-1 Quality Management	4.1.5 Accountability
Report on security compliance: access controls, integrity of the approval record	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10 Security by design; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance and quality data and driving improvements. Trends across production cycles are examined: are coherence problems decreasing, are fewer approval rounds needed, is the reuse of components rising, and is interpretability improving? Improvement opportunities are identified. Examples are refining indicator methods, improving coherence checks that produce false alarms, streamlining the approval workflow, improving visualizations, and reusing more validated outputs. Feedback from users through VS-10 is part of the input. Recommendations are fed back to the design sub-processes, prioritized by impact and feasibility, and tracked through the following cycles. This is SAF capability 4.1.4 *Improvement management* in practice.

Activity	Key outputs	SAF process	Other reference
Analyze quality trends across production cycles, including user feedback	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: indicator methods, coherence checks, approval, visualization, reuse	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Feed recommendations back to the design sub-processes or to VS-04 Estimation, Analysis, and Statistical Modeling	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-05 management feedback loop: products and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, coherence and interpretability reports to the quality manager, compliance reports to VS-07, component issues to VS-04, and improvement recommendations back to Design

Metadata and paradata per process type

In VS-05, metadata is what makes a composite product interpretable: each product must carry its components, methodology, coherence results, and approval. The table shows what each process type contributes. Data metadata describes the products. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Product specifications, composite-indicator definitions, coherence criteria, content templates, approval and confidentiality policy	Log of design decisions, methodology version, review and approval records
Implementation	Integration configurations, parameters of indicator construction and coherence checks, product templates	Build log, test results, coherence test outcomes, deployment approvals
Execution	Component lineage per product, composition and indicator method, coherence results, interpretive content, approval record	Operator identity, timing, software versions, trail of the approval workflow
Management	Coherence and interpretability assessments, product quality scores, lineage assessments, trend analyses	Coherence dashboards, product quality trends, improvement decisions, audit findings

PLACEHOLDER: VS-05 business object lifecycle: design artifacts govern implementation assets, which produce the pipeline that turns analytical Statistics (#4) into approved products (#4), generating component lineage, coherence, and approval records that management assesses and feeds back to design

5. Actors and roles

VS-05 brings subject-matter expertise, communication, and formal approval together. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the product manager and the coherence designer are ROL-06 Statistical designer, the indicator methodologist is ROL-07 Methodologist, the integration specialist is ROL-10 Data engineer, the subject-matter author is ROL-15 Subject-matter statistician, and the approval authority is ROL-17 Release manager. VS-04, which supplies the statistics, is a neighboring value stream and not an actor; the relation is governed by the handover interface in attribute 8. **[VC20]**

Actor or role	ID	Process type	Responsibility in VS-05 Statistical Product Composition
International statistical organization	ACT-07	External	Maintains product, indicator, and exchange standards and manuals, such as SDMX and index manuals
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-05 Statistical Product Composition
Data owner	ROL-02	Management	Accountable for approved versions of State #4; decides on internal access before release
Data steward	ROL-03	Execution	Registers approved products, maintains catalog entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, and retention of approved products before release
Statistical designer	ROL-06	Design	Designs the product portfolio, product specifications, coherence checks, and content templates
Methodologist	ROL-07	Design	Specifies composite indicators, indices, and the construction of time series
Metadata and data architect	ROL-08	Design	Designs the product metadata model and the propagation of component metadata

Actor or role	ID	Process type	Responsibility in VS-05 Statistical Product Composition
Data engineer	ROL-10	Implementation, Execution	Builds the composition components; selects and integrates components into the product structure
Platform engineer	ROL-11	Implementation	Configures composition services, connections to the output catalog, and access controls
Test manager	ROL-12	Implementation	Plans and carries out technical, coherence, and reproducibility testing
Subject-matter statistician	ROL-15	Execution	Runs coherence checks, explains discrepancies, and writes the analytical and interpretive content
Disclosure control specialist	ROL-16	Design, Execution	Assesses the confidentiality of products, including the risk of combining outputs
Release manager	ROL-17	Execution	Approves products against dissemination and confidentiality standards and signs them off for registration
Quality manager	ROL-18	Management	Monitors coherence and interpretability indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures confidentiality and standards compliance of products, and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-05 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-05 Statistical Product Composition
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the products and composite indicators, and build the composition pipeline

Capability group	Capabilities	Role in VS-05 Statistical Product Composition
1.3 Processing	1.3.6 Integrate statistics, 1.3.7 Statistical disclosure	Integrate outputs into products and check the confidentiality of combined outputs [**VC21**]
1.4 Analysis	1.4.1 Statistical (output) analysis	Check coherence and produce interpretive content [**VC21**]
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage product versions and metadata, assess coherence, integrate across registers and time
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain integration, indicator construction, and content tooling
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Check products against confidentiality standards and protect them before release
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward product approval, plan improvements, report on compliance

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-05 and which business functions they mainly use. It follows the mapping in *Business Processes, Mapping to value stream stages*. The activity tables in attribute 3 can name additional supporting processes. [**VC25**]

Stage	Realized by processes	Main functions
VS-05.1 Select and Integrate Source Datasets	PRD-5.1 Integrate data; PRD-6.1 Prepare draft outputs	FUN-3.4 Estimate and analyze
VS-05.2 Construct Composite Indicators or Indices	PRD-5.5 Derive new variables and units; PRD-5.7 Calculate aggregates	FUN-3.3 Process statistical data; FUN-3.4 Estimate and analyze
VS-05.3 Perform Consistency and Coherence Checks	PRD-6.2 Validate outputs	FUN-3.4 Estimate and analyze; FUN-5.2 Assure and review quality
VS-05.4 Generate Analytical and Interpretive Outputs	PRD-6.3 Interpret and explain outputs; PRD-7.2 Produce dissemination products	FUN-3.4 Estimate and analyze; FUN-3.6 Disseminate statistics
VS-05.5 Approve and Register Statistical Products	PRD-6.4 Apply disclosure control; PRD-6.5 Finalize outputs; DLM-3 Register and catalog data	FUN-3.4 Estimate and analyze; FUN-3.5 Protect statistical confidentiality; FUN-7.2 Manage metadata

PLACEHOLDER: ArchiMate view VS-05 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-05 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-1.3, PRD-2.1, PRD-2.5, DLM-1, DLM-4, OVA-1	Phase 1 Specify Needs (1.3); Phase 2 Design (2.1, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Implementation	PRD-3.2 to PRD-3.7, DLM-3, DLM-5	Phase 3 Build (3.2 to 3.7)	Capability Management	Data Analysis (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-6.1 to PRD-6.5, PRD-7.2 (content), DLM-2, DLM-3, DLM-5, DLM-6, OVA-1, OVA-2, OVA-4	Phase 6 Analyze (6.1 to 6.5); 7.2 Produce dissemination products (content only)	Production (execution)	Data Analysis (execution)	1.3.6 Integrate statistics, 1.3.7 Statistical disclosure, 1.4.1 Statistical (output) analysis
Management	OVA-1, OVA-2, OVA-4, DLM-5, DLM-7, DLM-8, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-05 most directly.

Principle	Statement	Application in VS-05 Statistical Product Composition
SAF-P01	Reuse before open source, open source before buying, buying before building	Composition, index, and visualization tools are reused across products first. Where new tooling is needed, open-source solutions developed with other NSIs come before purchase or own development. **[VC3]**
SAF-P07	Implement loosely coupled processes and systems	Composition is independent from estimation and dissemination. The approved product is the decoupling interface: dissemination uses it without knowing how it was composed.

Principle	Statement	Application in VS-05 Statistical Product Composition
SAF-P08	Data is shared property	Approved products and their components, once registered, are available to all authorized consumers. Composition reuses existing validated outputs instead of repeating analysis.
SAF-P09	No data without metadata and classification	Every product carries metadata: its component outputs and their accuracy, the composition and indicator method, the coherence results, the classification, and the approval record. This is what makes products interpretable and trustworthy.
SAF-P10	Security by design	Access controls, integrity verification, and a tamper-evident record of composition and approval are built into the pipeline, so every product can be attributed and reproduced. Products are protected against early disclosure.
SAF-P11	Privacy by design	Products are checked against confidentiality standards before registration. The disclosure risk of combining outputs is assessed by design, because combined figures can reveal what single figures do not.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 underpins interpretability: Design defines the product and the required metadata, Execution records component lineage, composition method, and coherence results, and Management assesses product quality. SAF-P10 governs the tamper-evident record of composition and approval across all stages. SAF-P11 applies in particular to *Design Approval, Confidentiality, and Versioning Framework* and to VS-05.5, where confidentiality and combination risk are confirmed before registration. SAF-P07 governs both the intake of analytical versions from VS-04 and the handover of approved products to VS-06.

8. Interfaces, services, and events

Boundary interfaces

VS-05 is loosely coupled (SAF-P07) to its neighbors through two boundary interfaces. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. Each interface exchanges a registered, immutable artifact plus its metadata, never internal working data. Neither side therefore needs knowledge of the other's internal processing.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-04 Estimation, Analysis, and Statistical Modeling → VS-05 Statistical Product Composition (intake)	Registered version of Statistics (#4, *analytical*)	Methodology and parameters, weights, accuracy measures, model diagnostics, record of disclosure control applied, version identifier, lineage to the processed source, classification	*Output Data Registered*	VS-04 Estimation, Analysis, and Statistical Modeling keeps custody of the analytical version; VS-05 Statistical Product Composition takes responsibility for composition on receipt of the event
VS-05 Statistical Product Composition → VS-06 Dissemination and Service Delivery (handover)	Registered approved product version (#4, *approved*)	Component lineage, composition and indicator method, propagated accuracy, coherence results, interpretive content, approval record, confidentiality status, product version identifier, classification	*Statistical Product Registered*	VS-05 Statistical Product Composition keeps custody of the product version; VS-06 Dissemination and Service Delivery takes responsibility for final disclosure control, release, and user support on receipt of the event

Business services

Service	Offered to	Realized in	Interfaces
Integration and composition service	Composition teams	VS-05.1 Select and Integrate Source Datasets	Integration and composition engines, version-fixed catalog connection
Indicator and index service	Composition teams, methodologists	VS-05.2 Construct Composite Indicators or Indices	Interface for composite construction, query interface for methodology and parameters
Coherence validation service	Quality managers, subject-matter statisticians	VS-05.3 Perform Consistency and Coherence Checks	Coherence reports and discrepancy lists
Analytical content service	Authors, editors, VS-06 Dissemination and Service Delivery	VS-05.4 Generate Analytical and Interpretive Outputs	Content templates, editorial workflow
Product provisioning service	VS-06 Dissemination and Service Delivery	VS-05.5 Approve and Register Statistical Products	Event notification (*Statistical Product Registered*) and a catalog query interface to find approved products, their lineage, and their accuracy

Service	Offered to	Realized in	Interfaces
Product catalog service	All data consumers in the NSI	Management sub-process *Manage Product and Version Lifecycle*	Search and browse interface for products, component lineage, coherence results, and approval records
Design and methodology service	Statistical designers, methodologists	All Design sub-processes of VS-05 Statistical Product Composition	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Output Data Registered	Availability of a component; start of VS-05.1 Select and Integrate Source Datasets when all components of a release are available (published by VS-04 Estimation, Analysis, and Statistical Modeling)
Coherence check failed	Return of a component to VS-04 Estimation, Analysis, and Statistical Modeling, or explanation of the discrepancy
Product approved / Product rejected	Outcome of approval in VS-05.5 Approve and Register Statistical Products
Statistical Product Registered	Completion of VS-05.5 Approve and Register Statistical Products; triggers VS-06 Dissemination and Service Delivery
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-05 event choreography between VS-04, the composition teams, the release manager, and process management across Design, Implementation, Execution (select and integrate, construct composites, check coherence, write content, approve and register), and Management (key performance indicators, coherence, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-05	Receives from VS-05
VS-07 Governance and Compliance	Approval and confidentiality policy, standards policy	Approval records, confidentiality and standards compliance evidence
VS-08 Innovation and Methodological Development	New indicators, index methods, and forms of visualization	Product gaps and improvement needs
VS-09 Infrastructure and Platform Enablement	Composition and content tooling, versioning store, catalog, event interfaces	Capacity requirements around release dates
VS-10 Customer and Stakeholder Engagement	User needs for products, breakdowns, and interpretation	Planned products and their content for consultation

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
On-time product rate	Share of products composed within the window of the release calendar	Per product, per cycle
Coherence check pass rate	Share of products that pass the coherence checks on the first run	Per product, per cycle
Number of approval rounds	Number of approval cycles before sign-off	Per product, per cycle
Component reuse rate	Share of products that reuse existing validated outputs	Per cycle
Reproducibility rate	Share of products that can be regenerated from components and method	Per cycle
Processing time	Time from the availability of all components to product registration	Per product, per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-05 are part of the SAF Toolkit.

Toolkit: SAF-TK-V05 VS-05 Statistical Product Composition companion [**VC17**]